PPS 11.2

① D

② B

③ B

④ $1.0\,mm^2 = 10^{-6}\,m^2$

$2.0\,W/m^2 \times 10^{-6}$

$= 2 \times 10^{-6}\,W$

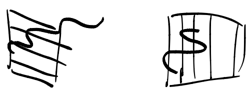
in 100s, 20×10^{-5} J tra

$= 2.0 \times 10^{-4}$

⑤ C

⑥ C

⑦ i

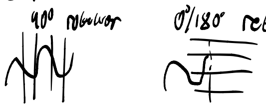


A polarized wave is one in which the direction of oscillation is aligned on one plane.

⑧ The transmitter emits a polarized signal.

The signal cannot go through the grille at first, because the direction of oscillation of the signal is perpendicular to the direction of the bars of the grille.

When the grille has rotated 90°, the signal can pass through fully, as the direction of oscillation is parallel to the gaps in the grille.



As grill rotates to 90°, more of the wave passes through the grile, so more of the signal is detected.

⑨ Using a polarized filter, face it towards the reflected ray and rotate it. If the light ray fully disappears at any point,

⑩ i) 4cm

ii) Intensity $= \frac{P}{A}$

As amplitude decreases, intensity decreases (since $I \propto A^2$)
This happens because energy is lost due to friction as the water particles slide over eachother.

As energy is lost due to friction, power decreases because $P = \frac{E}{t}$ (rate of energy transfer).

And since $I = \frac{P}{A}$, as P decreases I decreases.

⑪ a) unpolarized: light rays are oscillating in random direction

b) partially polarized

c) plane polarized.

⑫ By Malus' Law,

$I_t = I \cos^2\theta$

$I \propto A^2$

a) $I_1 = I_0 \cos^2 30$ $\quad A_1 = A_0 \cos 30$

$= \frac{3}{4} I_0$ $\quad = \frac{\sqrt{3}}{2} A_0$

b) $I_2 = I_0 \cos^2 45$ $\quad A_2 = A_0 \cos 45$

$= \frac{1}{2} I_0$ $\quad = \frac{1}{\sqrt{2}} A_0$

The remaining energy is transferred to the polarizing filter as heat.